

Homework Assignment 1

your name

2026-10-07

Getting Started: Download the .qmd template and datasets for this assignment from [this Google Drive folder](#). Save all files in the same folder before beginning.

Instructions: Complete all answers in this .qmd file and render it to a PDF to submit on Gradescope by 10/17 at 11:59 PM. For problems that involve a lot of math (e.g., derivations or proofs), you may handwrite your solution, take a photo, and insert the image instead. To insert an image, add a code chunk with the following line and replace "filename.jpg" with your image's filename, updating the file path if your image is not saved in the same folder as your .qmd:

```
knitr::include_graphics("filename.jpg")
```

-
1. (3 pt) When asked to state the simple linear regression model, a student wrote it as follows:

$$E(Y_i) = \beta_0 + \beta_1 X_i + \epsilon_i$$

Do you agree? Explain.

Answer:

2. Consider the simple linear regression model

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$$

(a) (3 pt) Assume that $X = 0$ is within the scope of the model. What is the implication for the regression function if $\beta_0 = 0$? Write the simplified regression equation and describe how the regression line would appear on a graph.

Answer:

(b) (4 pt) Under the assumption of $\beta_0 = 0$, derive the least squares estimate of β_1 .

Answer:

(c) (3 pt) How do you fit such a model using the `lm` function?

Answer:

```
# Your code here
```

3. Consider the simple linear regression model

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$$

(a) (3 pt) What is the implication for the regression function if $\beta_1 = 0$? Write the simplified regression equation and describe how the regression line would appear on a graph.

Answer:

(b) (4 pt) Under the assumption of $\beta_1 = 0$, derive the least squares estimate of β_0 .

Answer:

(c) (3 pt) How do you fit such a model using the `lm` function?

Answer:

```
# Your code here
```

4. (3 pt) The summation of residuals equals zero for the simple linear model. Does that imply the summation of random errors ϵ_i in the model equals zero? Does $E\left(\sum_{i=1}^n \epsilon_i\right) = 0$? (*Hint: think about the difference between residuals $\hat{\epsilon}_i = Y_i - \hat{Y}_i$ and the true errors $\epsilon_i = Y_i - E(Y_i)$.) Explain your reasoning.*

Answer:

5. (15 pt) Consider the simple linear regression model

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$$

(a) (10 pts) Show that minimizing the sum of squared residuals leads to the following least squares coefficient estimates:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2}$$

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$$

where $\bar{X} = \sum_{i=1}^n X_i/n$ and $\bar{Y} = \sum_{i=1}^n Y_i/n$.

(b) (5 pts) Show that $\hat{\beta}_0$ and $\hat{\beta}_1$ are unbiased estimators.

Answer:

6. GPA. The director of admissions of a small college selected 120 students at random from the new freshman class in a study to determine whether a student's grade point average (GPA) at the end of the freshman year can be predicted from the ACT test score.

```
# Load the GPA dataset
# gpa <- read.csv("path/to/gpa_data.csv")
```

(a) (3 pt) In this problem, what is the predictor variable, and what is the response variable?

Answer:

(b) (3 pt) Construct a scatter plot of the predictor and response variables. Comment on your impression of the relationship.

```
# Your code here
```

Answer:

(c) (3 pt) Fit a simple linear regression model using the `lm` function and report the estimated regression function.

```
# Your code here
```

Answer:

(d) (3 pt) Add the fitted line to the scatter plot in (b). Does the estimated regression function appear to fit the data well?

```
# Your code here
```

Answer:

(e) (3 pt) Obtain a point estimate of the mean freshman GPA for students with ACT test score equal to 30.

```
# Your code here
```

Answer:

(f) (3 pt) What is the point estimate of the change in the mean response when the entrance test score increases by one point?

Answer:

(g) (3 pt) Obtain the residuals and check whether they sum to zero.

```
# Your code here
```

Answer:

(h) (3 pt) Estimate σ^2 and σ . What are the units of them?

```
# Your code here
```

Answer:

7. Airfreight breakage. A substance used in biological and medical research is shipped by airfreight to users in cartons of 1,000 ampules. Data were collected on the number of times the carton was transferred from one aircraft to another over the shipment route and the number of ampules found to be broken upon arrival.

```
# Load the airfreight dataset
# airfreight <- read.csv("path/to/airfreight_data.csv")
```

(a) (6 pt) Obtain the estimated regression function. Plot the estimated regression function and the data. Does a linear regression function appear to give a good fit here?

```
# Your code here
```

Answer:

(b) (3 pt) Obtain a point estimate of the expected number of broken ampules when 1 transfer is made.

```
# Your code here
```

Answer:

(c) (3 pt) Estimate the increase in the expected number of ampules broken when there are 2 transfers as compared to 1 transfer.

Answer:

(d) (3 pt) Verify that your fitted regression line goes through the point $(\bar{X}, \bar{Y})$.

```
# Your code here
```

Answer: